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Non-business services performance forecasting for small urban areas using a spatiotemporal deep learning model

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ABSTRACT

Forecasting future economic outcomes is essential for effective policy and decision making in both the public and private sectors. However, the urban economic literature mainly focuses on causal relationships and neglects forecasting. In this research, in response to the lack of attention to essentiality of forecasting, the performance of non-business services is studied, and their outcomes are predicted for three years ahead. Given the complexity of relationships between social, environmental, and economic factors, a multivariate framework is used to forecast the three economic indicators of employment, business sales volume, and labor productivity. The spatiotemporal forecasting model is developed using Long Short-Term Memory (LSTM) Recurrent Neural Networks (RNN). It is trained and tested in Mecklenburg County, North Carolina, at the resolution of census block groups. The developed model is trained using historical data for the years 2002–2015 and tested using historical data for the years 2016–2018. The predictive performance of the models is judged to be high for all three outcome measures using the Mean Absolute Error (MAE) and normalized MAE metrics. The normalized MAEs for employment, sales volume, and labor productivity are 0.23, 0.21, and 0.16, respectively. This is more remarkable given the notorious difficulties of doing small-area economic forecasting.

1. Introduction

With cities producing 80 % of the global gross domestic product (World Bank, 2020), the study of urban economic growth and development is of great importance to maintain and guide the economic and social vitality of many developed and developing nations. In addition, the acceleration of changes in urban environments and the decentralization of housing and employment in the form of evolutions from monocentric to polycentric cities have led to fundamental shifts in the spatial distribution of economic activities (Kloosterman & Musterd, 2001). These changes and new dynamics make the study of urban economies increasingly important. In addition, urban economics has evolved from earlier exogenous growth models to endogenous growth models in which human capital and technology, as the most pivotal growth factors, are internal as opposed to external drivers. According to endogenous growth theories, these factors are the main determinants of sustained growth over the long term (Martin & Sunley, 1998). Urban economics literature covers the intricate interactions between its different elements, such as land use, transportation, land value, housing affordability, market dynamics, and location decisions of households and firms.

However, most of the research on urban economies has focused on the causal connections between its elements (Credit, 2018; John, 1996; Liu et al., 2011; Lv et al., 2021; Martínez & Viegas, 2009; Mohammad et al., 2013; Ryan, 1999; Schmidt et al., 2022; Shafik, 1994; Yarbrough, 2014), while only a small fraction has been devoted to predictions and future forecasts (Glaeser et al., 2018; Rajesh et al., 2021). In addition, the available literature predominantly uses conventional statistics and econometrics methods, lagging behind other disciplines in using state-of-the-art artificial intelligence (AI) methods such as deep learning. Only a few studies have applied new AI techniques in this research area in recent years (AlKhereibi et al., 2023; Arribas-Bel et al., 2021; Combes et al., 2022; Delmelle & Nilsson, 2021; Delmelle, 2022; Hatami et al., 2023; Fan et al., 2023; Ron-Ferguson et al., 2021). The current literature is further explained in detail in Sections 2.1 to 2.4. Furthermore, most of these studies are focused on urban growth models in large scale (Kim et al., 2023; Nwokolo et al., 2023; Tsagkis et al., 2023).

Understanding the variabilities in economic growth and

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development over time and space is of tremendous importance. Indeed, predictions of economic growth can tangibly serve to identify emerging urban districts that progress towards wholesale changes in the spatial structure of cities. Identification of these areas is very valuable for real estate investors, land developers, and business owners to find locations with better return opportunities in the future. In one study, Berg (2014) analyzed entrepreneurs' location decisions through interviews and found that they usually had limited information on these locations and their future returns. According to this study, most of the location decisions are made by chance. This dearth of information underscores the strong urge for having knowledge about future dynamics beforehand. On the other hand, future forecasts of economic growth can help to identify declining neighborhoods. Being able to anticipate neighborhoods that may be declining in the future can be very helpful for planning, policy making, and decision making in relation to transportation, housing and infrastructure, allocation of resources and services, jobs availability, etc., through the lens of spatial and social equity. In conclusion, future forecasting of economic growth sheds valuable light on future opportunities and threats, although such endeavors suffer from a lack of attention in urban economics literature. In this study, a spatiotemporal model is developed to forecast the business performance of neighborhoods for three years ahead. To this end, a deep learning model is proposed, trained and tested using historical data for the years 2002 to 2018.

Most studies of urban economics, whether in causal relationships or future predictions, are conducted at large scales, exploring the economic mechanisms in sets of cities or regions (Billings & Johnson, 2016). However, more granular analysis is often a better approach. Having an understanding of economic mechanisms at the neighborhood level is a decisive advantage owing to the prospects in guiding the location decision-making of firms and businesses.

This study aims to predict future economic outcomes of non-business services at the level of census block groups, representing neighborhoods in Mecklenburg County, NC, in the United States. Neighborhood economic outcomes are measured by multiple indicators, including employment, business sales volume, and labor productivity as an economic indicator derived from the first two variables.

Given the known interdependencies of sociodemographic, economic, and built environment factors in urban economies, multivariate forecasting is used to predict each neighborhood's productivity. In this predictive model, potentially related covariates such as built environment characteristics, agglomeration forces, business characteristics of the neighborhoods, and real estate investments are taken into account as covariates.

Built environment characteristics have been hypothesized to be associated with the economic development of cities at different geographic levels, such as neighborhoods, particularly the economic development of non-business services such as retail. A range of urban planning and design theoretical approaches have been developed to enhance the transit accessibility and economic vitality of neighborhoods, such as new urbanism, mixed-use development, Transit-Oriented Development (TOD), and smart growth (Day, 2003; Hatami & Thill, 2022; Hess & Lombardi, 2004; Krueger & Gibbs, 2008). In these approaches, built environment characteristics such as transit proximity, Central business district (CBD) proximity, density, jobs-housing ratio, and land-use mix have been promoted for their positive impact on economic development. Given these studied impacts, the built environment characteristics are incorporated in the predictive model of this study as potential predictors.

Additionally, attributes of other businesses such as their sales volume, count of employees, and business type (e.g. headquarter, branch, single location, etc.,) are incorporated in the model to capture agglomeration forces such as specialization, urbanization, and location quotient. Furthermore, trade area demographics such as income, race, and age are incorporated into the model as they are expected to have an impact on the neighborhood's economic performance (Kumar & Karande, 2000). Lastly, real estate investments have been known as leading indicators of the economy (Manuele, 2009; Strauss, 2013). An increase in the number of new residential construction or commercial developments can inform us about economic development in advance. Therefore, building permits of different types, such as new residential and commercial constructions and their renovations, are used as leading predictors.

Annual historical data of the years 2002 to 2018 are used for the economic outcomes and their covariates. Given the complexity of relationships between economic predictors and urban dynamics components, and given the limitations in assumptions of traditional econometrics such as linearity, deep learning (DL) is used for the time series forecasting as it is capable of handling the anticipated complexities of the empirical problem under study. Due to the high performance of machine learning (ML) and DL models in predicting socioeconomic dynamics, they have been more widely used in recent years (Arribas-Bel et al., 2021; Delmelle & Nilsson, 2021; Glaeser et al., 2018; Hatami et al., 2023; Rajesh et al., 2021; Reades et al., 2019; Ron-Ferguson et al., 2021), instead of conventional econometric models. In socioeconomic phenomena, there are complex relationships between a large number of features. DL models digest these relationships in high-dimensionality data very well and with few or no limitation. In contrast, in econometric models, assumptions have to be met and data engineering needs to be done carefully.

2. Literature review

There is a large body of literature on urban economies. The literature investigates this multifaceted topic from different dimensions, such as land-use/transportation integration, neighborhood change, and sustainability. Most of these studies explore the associations and causal relationships between various aspects of urban dynamics rather than forecasting over a certain planning horizon. Examples include studies investigating the impact of transportation on property values (John, 1996; Martínez & Viegas, 2009; Mohammad et al., 2013; Ryan, 1999; Schmidt et al., 2022), the impact of transportation on retail commercial development (Credit, 2018; Yarbrough, 2014), identifying the determinants of polycentric structures (Liu et al., 2011; Lv et al., 2021), and relationships between economic development and environmental issues (Shafik, 1994). Less attention has been paid to future trends predictions.

2.1. Artificial intelligence and data-driven methodologies in studying urban economies

In addition, the growth in high-resolution data availability and recent advancements in data-oriented approaches that originated in computer science, such as machine learning and deep learning, are expected to bring considerable advancements in studies of urban economies and economic geography. However, it is only recently that serious attention has turned to the applications of artificial intelligence (AI), ML, and DL to problems in these domains of application. Delmelle (2022) conducted a review on recent applications of data-driven and ML methods for predictive and explanatory analysis on neighborhood change processes. Hatami et al. (2023) utilized ML to predict the commuting modal split in Mecklenburg County, NC. They further applied the SHAP ML method to investigate the impact of the built environment on commuting mode share. The ML methods used in this paper resolve complexities such as non-linear associations that conventional econometric models are not able to capture. Delmelle and Nilsson (2021) performed text analysis on real estate advertisements using ML to explore the relationships between advertisements text data and neighborhood socioeconomic characteristics. Their trained ML model performed well in predicting the neighborhood type. This finding indicates the potential of using AI and text analytics for predicting future neighborhood changes before they happen. Arribas-Bel et al. (2021)

used the Approximate DBSCAN method to delineate urban areas in Spain using data from 1 million residential and non-residential buildings. Then, they compared their delineated boundaries with the commuting-based boundaries and administrative boundaries and found more consistencies with the commuting-based boundaries. Emphasizing the important role of historical data in some research areas of urban economies such as agricultural productivity and urbanization; urban growth, city structure and transportation; communication, transportation and trade; and migration and interregional mobility, Combes et al. (2022) studied how machine learning methods including the random forest and neural networks can be applied for the treatment of historical data. Kourtit et al. (2021) studied global cities' safety and security, and economic development potentials, namely magnetism. They used variables related to the stated dimensions. Using an advanced sequential cluster dynamics analysis method, they identified clusters of cities. They found that cities' safety and security were important predictors of their economic performance. Ron-Ferguson et al. (2021) used the random forest machine learning technique to study urban development and change through demolition and new construction activities. Applying ML, Reades et al. (2019) studied neighborhood change from 2001 to 2011 to predict gentrification in London neighborhoods. Overall, most of the applications of AI in economic geography are very recent and not much attention has been paid to this until very recently.

2.2. Fine spatial granularity in urban economics studies

In the literature on urban economies, studies on future forecasts have usually been on predictions of population (Booth, 2006) in large scale and coarse resolutions or urban growth and land use change using satellite imagery datasets (Aburas et al., 2016; Allen & Lu, 2003; Batty & Torrens, 2001; Clarke & Gaydos, 1998; Park et al., 2011; Triantakonstantis & Mountrakis, 2012), not socioeconomic data. Only a small number of studies have investigated forecasts and predictions of economic growth with actual economic indicators using socioeconomic data. Pentland (2020) studied the interactions between communities and flows of ideas between them as an important predictor of economic growth outcomes. He found that diversity of ideas and interactions between agents at individual, business and national levels played an important role in predicting economic growth. Chong et al. (2020) predicted economic prosperity and growth at the neighborhood level in Istanbul, Turkey, Beijing, China, and several cities in the United States. They used the diversity of consumed goods and services within neighborhoods and interactions and flows between them as predictors of economic growth and prosperity. Wei (2018) predicted economic growth in the presence and absence of high-speed rail in Changsha City, China, to evaluate the impact of high-speed rail on economic growth. Applying regression and Gray prediction models, they used data from 2001 to 2009 and predicted the economic growth from 2010 to 2015. Using data from remote sensing and interviews, Warth et al. (2020) predicted the socioeconomic indicators, including income, building assets, and educational attainment, to support future infrastructure planning.

2.3. Economic heterogeneity across industries

Urban economies have largely been explained through agglomeration economies or economies of scale, which varies significantly over industries (Giuliano et al., 2019; Hanlon & Miscio, 2014). Economies of scale mean that firms and businesses agglomerate and cluster together in the urban space as they gain benefits from this agglomeration depending on their industries and input-output linkages between them and other industries. This scaling effect is a result of various forces such as input-output linkages between industries, common access to the same pool of labor between firms in the same industry, knowledge spillover effects between firms in similar or different industries in close proximity, natural advantage including factors such as mutual access to transportation infrastructure, service providers and consumers (Puga, 2010). Having access to common resources between firms or businesses through the mentioned forces reduces costs such as costs of transportation, labor, inputs such as natural resources, and warehousing and distribution. The discussed agglomeration forces are expected to play an important role as predictors of economic outcomes.

As mentioned earlier agglomeration forces and their benefits vary by industry. Much of these variations occur between industries, whether they are in the group of business services or the group of non-business services. The biggest difference between these two groups is their demand side. Business services provide services to other businesses, while non-business services provide goods and services for individuals as customers. As a result, their input-output linkages, labor force skill level, sales volume, production scale, sensitivity to changes in other industries, and many other factors are different. Given these fundamental differences, it is important to study these industries in separate groups.

Although the United States has been experiencing upward trends in the growth of business services in the economy as a result of tremendous advances in technology, non-business services still make up a considerable share of employment (56.57 % of total employment in the US in 2017) (US Census Bureau, 2017). Also, they comprise 54.6 % of total firms in the US. Additionally, non-business services provide jobs to low-skilled people with lower wages (only 0.38 % of the total US payroll while covering 56.57 % of total employees in 2017) (US Census Bureau, 2017). Given the accelerating growth in business services and high-skilled jobs, it becomes more important to direct attention to jobs for low-skilled individuals, particularly in growing cities such as Charlotte, North Carolina.

As mentioned earlier, factors affecting non-business services are different from business services. Non-business services depend on foot traffic. As a result, accessibility to these services plays an important role in their growth. Accessibility matters both for customers as service consumers and for workers as service providers. Higher accessibility for workers improves labor pooling and matching as one of the significant agglomeration forces (Chatman & Noland, 2011). Therefore, it is expected that areas with better transportation accessibility provide more opportunities for the success of non-business services. These accessibility factors include public transit such as light rail, parking availability, the walkability of the area, and accessibility by different modes of transit. Their demand side can be measured by their respective housing market, housing density, trade area, and spending power of neighborhoods in their service area, such as median income. Their supply side also plays an important role, such as accessibility to suppliers and to warehousing.

Despite the presence of heterogeneity over industries in urban economics, a majority of the forecasting literature, including the ones discussed above, has been on economic growth in general and not on specific industries like non-business services. Non-business services such as retail, restaurant and other services to non-business clients have been known as important drivers of economic growth or decline at the neighborhood level in the field of community development (Koebel, 2014; Zukin et al., 2009). The contribution of these services to the neighborhood and city economy has been one of the main focal areas of newer theoretical planning approaches such as new urbanism, Transit-Oriented Development, and smart growth. A number of studies have been dedicated to the relationships between the growth of these industries and neighborhood economic development. Waxman (1999) studied the revitalization of inner-city neighborhood business districts through economic theories of retail and commercial revitalization in Dorchester, Massachusetts. Similarly, Koebel (2014) studied neighborhood revitalization through commercial revitalization and growth in the number of retail stores and service establishments. This study argued that most of the research focus has been on the impacts of large-scale commercial developments, while overlooking the importance of disadvantaged neighborhood commercial districts and their influence on the vitality of neighborhoods. The study stated that growth or decline in the number of retail stores and service establishments was a function of market factors, including trade area demographics and demand characteristics, and non-market factors, including discrimination as a result of race and ethnicity. Dolega and Lord (2020) investigated the economic growth and decline of neighborhoods by dynamic trajectories of retail business performances between the years 2014 to 2016 in the Liverpool, UK, city region. They found that the trends of retail economic growth varied over space in this region. Focusing on grocery retailing, Pothukuchi (2005) investigated initiatives for increasing grocery investment and found variations in those initiatives in different sites. They found that systematic initiatives for the entire city were not common and recommended that planners increase investments in retail groceries in disadvantaged areas to promote economic development.

2.4. Forecasting non-business services

Recently, more attention has been paid to future forecasts of non-business services in urban economies. Using Yelp social media data, (Glaeser et al., 2018) studied the relationship between the number of restaurants, bars, cafes, and grocery stores, and gentrification in New York City zip codes and found a significant association. They found that local businesses could predict an increase in housing prices and gentrification as leading indicators. Rajesh et al. (2021) measured the resiliency of the retail industry of urban centers using five resilience indicators of the vacancy rate, retail turnover dynamics, growing sales history tenant structure, and economic crises from 2010 to 2019. On this basis, the study predicted the resiliency of the retail industry in urban centers in 2020 using gray prediction and moving probability-based Markov models.

3. Research design, methods, and data

3.1. Forecasting framework

This study proposes a framework to forecast the performance of non-business services at a fine spatial resolution (Fig. 1). Economic performance is measured by several indicators that complement each other, each one capturing a specific perspective on economic performance locally. The first economic indicator is employment. Employment trends in non-business services are essential to investigate as these businesses mainly employ low-skilled labor. In developed countries, high-skill jobs such as business service activities are increasingly taking a more important part in the economy. Therefore, as the low skilled workforce is at risk of being cut back and displaced by technology, it is essential to pay attention to employment in businesses with low-skilled labor. The second indicator is the business sales volume. This variable is selected as sales of non-business services such as retailing indicate the spending power of consumers and demands. Finally, the last economic indicator is the average sales volume per employee, which tracks businesses' productivity. Generated using the first two variables, the latter indicator is of great importance as it shows the profitability of businesses. Knowing about the future aggregate profitability of a neighborhood is critical from the businesses' perspective as it helps them to locate in or relocate to areas with higher returns.

As a result of the complexity of socioeconomic phenomena at fine spatial resolution and of the interdependencies between covariates, multivariate forecasting is used. The high dimensionality of covariates is collapsed into classes of business characteristics, demography of residents and workers, built environment characteristics, and real estate investments. Due to the forces of agglomeration economies, characteristics of businesses within and outside of the studied industry are

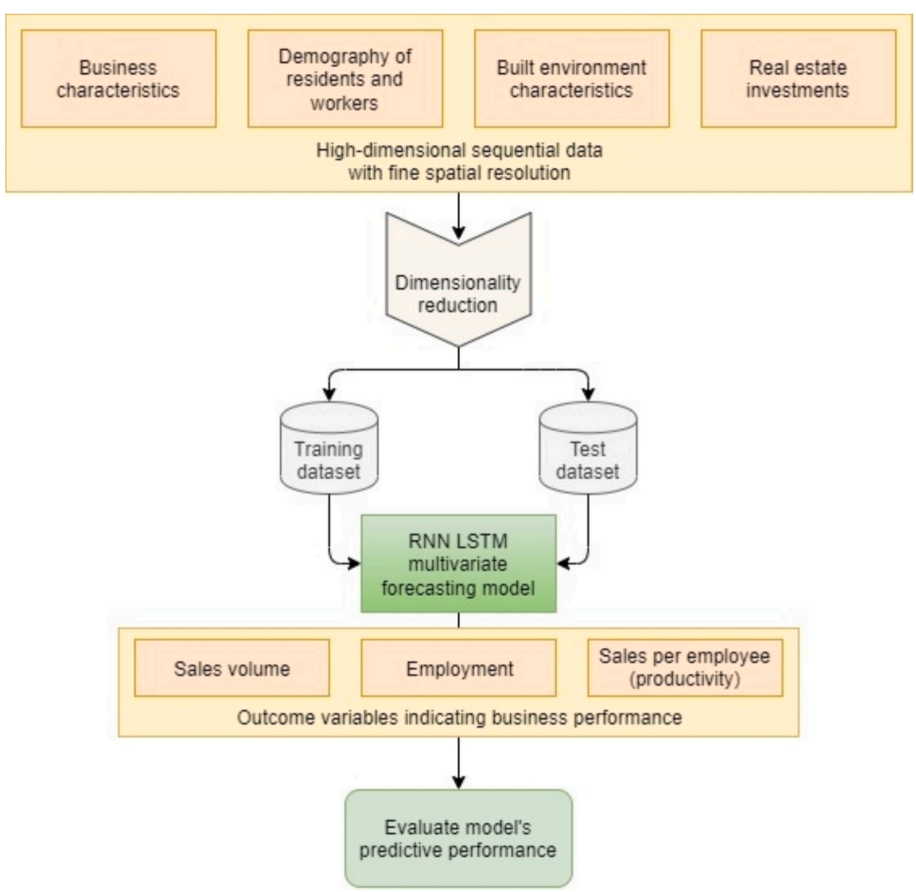

Fig. 1. Proposed framework for forecasting business performance of neighborhoods using an RNN LSTM multivariate model.

expected to have an impact on business performance. The demographic characteristics of residents and workers are expected to be strong predictors as they show the characteristics of consumers and of the labor pool, which link up to the demand and supply of the activities being forecasted. Since non-business services are dependent on foot traffic, built environment factors such as accessibility play an important role in the neighborhood's business performance. Lastly, real estate investments are used as a leading indicator of the economy.

Due to the complexity of the subject matter, our framework proposes to use DL methods to identify the combination of predictors that most effectively forecasts neighborhood economic performance. DL methods are capable of digesting complex situations in which there are non-linear relationships between covariates, when there is a large number of interdependent features, the amount of data is large, or when there are multiple seasonality and trends in the time series data. Another benefit of using the DL method over traditional econometrics and machine learning is that it requires less domain expertise, making it useful across a range of disciplines. In addition, agent-based modeling is another common method in urban dynamics forecasting. However, there are some limitations in these models and other simulation-based models. All of them have a rigid set of assumptions that simplify the phenomenon. They are extremely reliant on and sensitive to the model's input variables. The prediction models will also be univariate rather than multivariate. In contrast, cutting-edge data-driven methodologies, such as DL, draw knowledge from various aspects of the phenomena captured in the data. For the purposes of neighborhood economic performance forecasting, a multi-output multistep multivariate time series forecasting approach is proposed that is based on the Long Short-Term Memory (LSTM) Recurrent Neural Networks (RNN). In conclusion, the proposed advanced method of LSTM-RNN is more complicated to implement and takes more computational time than conventional methods in this discipline; however, it resolves a number of shortcomings as discussed above.

As an extension of artificial neural networks, deep neural networks are highly capable of capturing complexities due to having a large number of hidden layers in their architecture. An artificial neural network is composed of an input layer, an output layer, and a hidden layer(s). Neurons in the mentioned layers are connected through some links associated with weights and biases. Through the learning process, the model finds the optimized values of these parameters by backpropagation. A deeper network with a higher number of hidden layers can capture more features and complexities in data (Schmidhuber, 2015). As a result of the mentioned power of deep neural networks, along with the availability of high-performance computers, these models have become widespread in a variety of application areas, including in urban economics (Grekousis, 2019; He et al., 2018; Ma et al., 2023; Zheng et al., 2023). Furthermore, as a subset of DL methods, RNNs can digest complex relationships in sequential data. In RNNs, the output of the hidden layer at each time is saved and fed back to the layer while keeping a hidden state. Fig. 2 shows a simple illustration of the RNN architecture. U, W, and V are model parameters that are optimized through the training process. As shown in this figure and in eq. 1, the hidden layer receives information from both the input layer and earlier stages of itself. h_t is the new state; f_c is the function with parameter c; h_{t-1} is the old state; and x_t is the input vector at time step t. Given the high performance of RNNs in analyzing sequential data, they have received

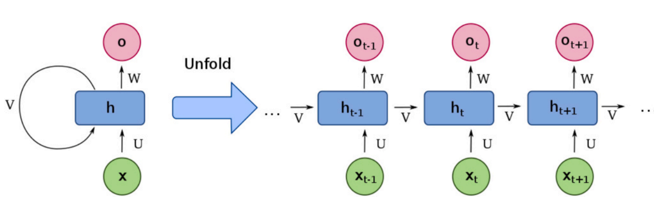

Fig. 2. Simple illustration of RNN.

great attention in the literature of different research areas, including urban analytics (Khusni et al., 2020; Nikparvar et al., 2021; Pan et al., 2022; Zhene et al., 2018). Moreover, the LSTM model enhances the model's performance since it resolves the vanishing/exploding gradient issue in RNNs. The LSTM model learns long-term dependencies and resolves the gradient-related issues through the cell state. The cell state is regulated by gates for removing or adding information (Graves & Graves, 2012).

$$h_t = F_c(h_{t-1}, x_t) \tag{1}$$

The data are split into training and test sets, with 80 % and 20 % proportions respectively, to fit the proposed model and evaluate its performance. The selected proportions provide us with a sufficient historical length for training and three years for the assessing of the model's performance. With increasing availability of longer histories of data in the future, larger proportions (say 30 %) can be used for testing purposes (further discussed in Section 5). The model's predictive performance is evaluated using the mean absolute error metric. Before training the forecasting model, due to the large number of features, dimensionality reduction is performed. The model is developed in Python using the Keras library (Chollet, 2015).

3.2. Study area

As a case study, Mecklenburg County, NC, is selected and forecasts are developed for geographic neighborhoods, which are taken to be census block groups. As the most populous county in NC until 2020 and with the city of Charlotte as its seat, Mecklenburg County has been experiencing fast population growth and accelerated economic development over the past decades, which makes it a well-suited area of study.

The county's population increased 58 %, from 700,458 in 2000 (U.S. Census Bureau, 2021a) to an estimated population of 1,110,356 in 2019 (U.S. Census Bureau, 2021b). Total employment has increased by 42 % from 2002 to 2018 (U.S. Census Bureau, 2021c).

The three industries of Retail Trade, Accommodation and Food Services, and Other Services studied in the second part of this study comprised 21.29 % of the total employment in Mecklenburg County in 2018, with the Retail Trade and Accommodation and Food Services in the top 5 industries among all 2-digit NAICS codes in terms of share of employment. These three industries had significant growth in Mecklenburg County from 2002 to 2018, with percentage changes of 35 %, 85 %, and 33 %, respectively (U.S. Census Bureau, 2021c).

Anticipating the continued growth of population and of the local economy, some plans were developed for this county to better integrate land use, urban design, and transportation systems, such as the 2025 Transit/Land-Use Plan, which was finalized in 1992. This plan entails five transit corridors and developments along them; the first corridor started operating in 2007 as the Blue Line of the Lynx light rail service. The establishment of this light rail line has led to developments incorporating high density, high diversity, low distance to transit, high destination accessibility, and careful design (known as the 5D design elements), including transit-oriented development projects in proximity to the rail transit stations. These developments place a compact mixture of residential, office, retail, and consumer services along the transit corridors. In this study, the annual historical data for the years between 2002 and 2018 are used to forecast the future years' economic outcomes and test the model's predictive performance.

3.3. Data sources

Data on sales volume and employment in the three studied non-business services, which are used as indicators of the neighborhoods' business performance, are collected from the US Businesses dataset from the data source Data Axle (Reference Solutions) (Data Axle, 2021). Sales volume is adjusted for inflation in the data source. Employment and sales volume data are reported by year from 2000 to 2018 for each business. Then, data are aggregated to the census block group boundaries for each industry based on the reported NAICS code of the business and for each year. Also, the count of businesses by industry is calculated for each census block group and year from Data Axle's US Businesses dataset.

In addition, data for the count of employees by industry, residents' and workers' income, age, educational attainment, race, or ethnicity are sourced from the Longitudinal Employer-Household Dynamics (U.S. Census Bureau, 2021c) dataset. These data are available in the census blocks for each year between 2002 and 2018. The census block data are aggregated to the census block group boundaries for each year. Data for capturing real estate investments are collected from the building permits dataset Basic Report for Posse Permits from the Mecklenburg County Open Data (2021). This dataset includes all issued building permits in Mecklenburg County for different categories, such as new residential constructions, new commercial constructions, residential and commercial renovations, and demolitions. Since real estate investments and building permits are considered leading indicators of the economy, the building permits data are collected with a one-year lag (Strauss, 2013). In other words, the study period in this research is 2002 to 2018, and the building permits are incorporated in the model for the years 2001 to 2017. Furthermore, variables representing the built environment, such as proximity to rail stations or proximity to the Charlotte CBD, are calculated from GIS shapefile data collected from (Mecklenburg County GIS Center, 2021). The proximity measures are calculated using the centroid of block groups.

3.4. Data description and processing

In this study, the economic performance of neighborhoods is forecasted for specific non-business services for future years. These industries include retailing, accommodation and food services, and other services with 2-digit North American Industry Classification System (NAICS) codes of 44–45, 72, and 81. These three industries are selected due to their close similarities from different aspects. They all provide services to individuals and are dependent on population foot traffic. They often operate with low-skilled labor. They are dependent on the spending power of residents in proximity to the points of sale and service delivery. Due to their dependence on foot traffic, accessibility measures such as transit proximity and CBD proximity are expected to have an impact on their performance. The historical data from 2002 to 2018 are used with an annual temporal resolution. The spatial resolution is that of census block groups representing neighborhoods.

As discussed in the forecasting framework section, three outcome variables are developed and used to predict the business performance of neighborhoods. Descriptive statistics of these three indicators are shown in Table 1.

Covariates that cover forces of agglomeration economies, including colocation using the number of businesses in the same industry, number of employed residents of each census block group indicating the labor pooling, and location quotient or localization are used. The location quotient is an index of a block group's specialization in non-business services relative to that of the entire county. It is calculated as the ratio of non-business services to all businesses in a block group divided by the same ratio in the entire county. In addition, characteristics of businesses, including the average firm size and average firm age, are included. Built environment variables, including CBD proximity, proximity to light rail, and jobs-housing ratio, are incorporated as other potential predictors. The CBD proximity of each block group is calculated as the distance between the CBD and the block group. Proximity to light rail is taken as the distance between each block group and its closest transit station following a straight line. Sociodemographic variables, including race, education attainment, the mean per capita income of workers and residents, and the mean age of workers and residents, are added as control variables. Lastly, a binary variable indicating the economic recession in 2009 and 2010 is added to the forecasting model

Table 2
List of covariates used in the multivariate forecasting model (census block group aggregates by year).

Covariate category	Covariate	Data source
Business characteristics	Count of employees by industry	Data Axle, 2021
	Sales volume by industry	Data Axle, 2021
	Count of businesses by type (4 categories of headquarters, branch, independent, and individual) by industry	Data Axle, 2021
	Location quotient of businesses by industry	Data Axle, 2021
Demography of residents and workers	Count of employees and residents by age group (29 or younger, 30–54, 55 or older)	U.S. Census Bureau, 2021c
	Count of employees and residents by income group (earnings $1250/month or less, $1251/month to $3333/month, and greater than $3333/month)	U.S. Census Bureau, 2021c
	Count of employees and residents by race group (White, Black, American Indian, Asian, Hawaiian)	U.S. Census Bureau, 2021c
	Count of employees and residents by ethnicity group (Hispanic, non-Hispanic)	U.S. Census Bureau, 2021c
	Count of employees and residents by educational attainment group (less than high school, high school, some college education, bachelor's degree or advanced degree)	U.S. Census Bureau, 2021c
	Count of employees and residents by sex group (male, female)	U.S. Census Bureau, 2021c
	Count of employees by firm age group (0–1, 1–2, 2–3, 4–5, 6–10, and 11+)	U.S. Census Bureau, 2021c
	Count of employees by firm size group (0–19, 20–49, 50–249, 250–499, and 500+)	U.S. Census Bureau, 2021c
Built environment characteristics	Distance to the closest public transit station	Mecklenburg County's GIS Center (2021)
	Distance to CBD	Calculated in ArcGIS
	Jobs-housing ratio	U.S. Census Bureau, 2021c
Real estate investments	Count of building permits for demolitions	Mecklenburg County Open Data (2021)
	Count of building permits for residential new construction	Mecklenburg County Open Data (2021)
	Count of building permits for commercial new construction	Mecklenburg County Open Data (2021)
	Count of building permits for residential renovations	Mecklenburg County Open Data (2021)
	Count of building permits for commercial renovations	Mecklenburg County Open Data (2021)

Table 1
Descriptive statistics of outcome variables (observed values in all years and all block groups).

Outcome variable	Count	Mean	Standard Deviation	Min	Median	Max
Employment (person)	9435	223.31	430.32	0	75	533
Sales volume ($1000)	9435	38,000.21	74,805.64	0	10,894.33	718,762
Sales per employee ($1000/person)	9435	119.01	74.32	0	107.04	743.44

and serves as a fixed effect on neighborhood performance. Table 2 shows a detailed list of all covariates used in this paper. Due to the large number of variables (300+ variables for 17 years), their summary statistics are not published in this article.

Data are split into training-validation and testing sets with 80 %–20 % shares, respectively. This study predicts the last three years of data as 2016, 2017, and 2018 as the 20 % split to test the model performance. Summary statistics of outcome variables for the three forecasting years are shown in Table 3. In this study, there are more than 300 features (predictive variables). As a result, principal components analysis (PCA) is applied on covariates (for years 2002–2015), as a powerful dimensionality reduction technique. The number of components is set to four as they collectively explain 98 % of the cumulative variance. The PCA reduces the runtime tremendously (from more than 12 h to about 3 h on a PC with 32 GB RAM, 8 core CPU and GPU) as our dataset has more than 300 features. It also improved the model performance significantly.

To improve the model performance, this study employed a data augmentation technique by training the model on data of all block groups together, instead of training and predicting each block group individually. It used a window size of 6 years for the training part. This window size is chosen based on our investigations of temporal dependence between the data. Data is highly correlated for up to 6 years prior. After training the LSTM-RNN model, forecasting is done for the next 3 years for each census block group individually. Out-of-sample validation is done using the remaining 20 % of the data. The mean absolute error value is used for model performance evaluation. Lastly, in the first set of results, there was a small number (less than 10) of negative predictions close to zero for employment and sales volume variables, which is a problem commonly encountered in time series forecasting (Burkom et al., 2007). In order to avoid negative predictions, as counts cannot be negative, the logarithm of the outcome variables was taken before training the model, and then, predictions were made using the exponential function. Furthermore, the spatial lag effect was tested in the models by aggregating the values of neighboring units for all variables related to the residential area characteristics (RAC data from the LEHD dataset). Results did not show improvements in fit in comparison with the non-spatial models and are therefore not presented in detail hereunder. One potential reason is the fine spatial resolution of the study. Block groups share similarities with their neighboring units due to their small size in relation to the spatial scope of the study.

4. Results

In this study, three economic indicators of local economic performance that involve employment, business sales volume, and labor productivity, respectively, are forecasted in Mecklenburg County census block groups for the years 2016, 2017, and 2018.

The model predictive performance is evaluated for all three outcome variables by the mean absolute error (MAE). The MAE is reported as an average over the three forecast years, and also for each forecast year individually. Given the variable range of observed values across block groups, the average MAE is normalized by both the mean and range of observed values to better understand the errors in comparison with observed values. The normalized MAE metrics are used as they provide us with information about the magnitude of the error relative to the observed values. They are more informative as there is a wide range of observed values across all block groups. In addition, this metric is used instead of percentage error, which is one of the most common relative error measures, since the practical magnitude of the percentage error depends on the range of observed values as well. Therefore, normalized error metrics are used to evaluate the predictive performance of the models. As indicated in Table 4, for all three outcome variables, the MAE values normalized by range are less than 0.04 for each of the three predictive years. This shows the high predictive performance of the proposed model. The MAEs normalized by mean have values of 0.23, 0.21, and 0.16 for employment, sales volume, and labor productivity, respectively. Taken together, these values show that our model is fully capable of predicting the neighborhood business performance with a high accuracy, that is 97 % and higher when compared with the range values, and an accuracy of 76 % and higher when compared with the mean values. In addition, the table shows the variability of the performance metrics over the years. The model performs better for the first and second years, in comparison with the third year, as is typical of forecasting over an extended planning horizon. The statistical distribution of errors is also examined (2016 Normalized Absolute Errors (NAE)

Table 4
Forecasting model performance evaluation using Mean Absolute Error (MAE) and Normalized Mean Absolute Error (NMAE) (normalized by the observed mean and by the observed range), by forecast year.

Evaluation metric	Forecast year	Predicted variable		
		Employment	Sales volume	Average sales per employee
MAE	All years	61.846	8782.548	20.69
NMAE (by mean)	All years	0.235	0.206	0.164
NMAE (by range)	All years	0.011	0.012	0.034
MAE	2016	35.983	8782.548	14.544
MAE	2017	37.998	7370.391	13.341
MAE	2018	111.558	11,956.066	34.185
NMAE (by mean)	2016	0.135	0.187	0.107
NMAE (by mean)	2017	0.138	0.155	0.096
NMAE (by mean)	2018	0.452	0.358	0.327
NMAE (by range)	2016	0.007	0.012	0.025
NMAE (by range)	2017	0.007	0.011	0.022
NMAE (by range)	2018	0.031	0.023	0.084

Table 3
Descriptive statistics of outcome variables (2016, 2017, 2018 average observed values).

Outcome variable	Year	Count	Mean	Std	Min	Median	Max
Employment (count)	Three-year average	1665	262.51	481.19	0	98.67	5333
	2016	555	266.03	515.64	0	84.67	5305.33
	2017	555	274.89	528.27	0	90.33	5333
	2018	555	246.61	387.45	0	116	3531.67
Sales volume ($1000)	Three-year average	1665	42,643.60	81,918.40	0	12,391	701,299
	2016	555	47,051.80	90,148.60	0	13,599	701,299
	2017	555	47,509.40	90,011.50	0	14,024.30	683,187
	2018	555	33,369.50	61,627.10	110.67	10,471	513,898
Sales per employee ($1000/employee)	Three-year average	1665	126.39	82.91	0	112.21	600
	2016	555	135.94	86.92	0	123.68	588.19
	2017	555	138.75	89.83	0	125.38	600
	2018	555	104.50	65.54	4.57	94.74	403.89

in Fig. 3) to pinpoint cases where the model underperforms. The figure shows that most block groups have very small error values, close to zero. The thin tail of the distributions indicates that only a very small number of block groups have larger errors. In short, not only the model performs well in general, but outliers are scarce.

Specifically, Fig. 3 shows the statistical distribution of normalized predictive errors (normalized by the range of observed values) for each of the outcome variables for the year 2016. Comparing the histograms of the three outcome variables shows that, for labor productivity, the errors are more skewed and there are more block groups in the larger error classes. This variable is calculated based on the other two variables (employment and sales). As a result, the level of uncertainty and noise is increased in the productivity metric. Therefore, predictions become somewhat more challenging.

Having established the forecasting method performs very well overall for the study area, this paper can proceed with a closer examination of local patterns of forecasts across Mecklenburg County. In Fig. 4, the observed values (on the left) and predictive errors (on the right) are illustrated using NAEs (normalized by the range of observed values) for the three outcomes variables of (a) employment, (b) business sales volume, and (c) productivity given by sales per employee in 2016. The mapping is done with an equal interval classification method. The absolute errors normalized by the mean of observed values exhibit very similar patterns in their spatial distribution to those of the absolute errors normalized by range, and for the sake of brevity are left out. As shown in Fig. 4, most block groups have very low predictive errors (first class), that is under 0.066, 0.009, and 0.090, for employment, sales volume, and labor productivity, respectively. Only a few block groups (1 %, 1 %, and 2 % of block groups for employment, sales volume, and labor productivity, respectively) have higher errors for all outcome variables. The maps of observed values show that these block groups with higher normalized errors all have larger observed values. It is natural to have larger errors in predicting larger values. Since these errors are normalized by a constant value (average observed values), the normalized errors naturally become larger. Even these block groups with higher errors actually have normalized errors under 0.33, 0.02, and 0.22 for employment, sales volume, and labor productivity, which shows performance above 67 %. Furthermore, spatial dependence is tested on errors using the Global Moran's *I* statistic (Moran, 1950). The spatial weight matrix is built using the inverse distance specification. Results show that there is no spatial autocorrelation with index values of 0.035, 0.064, and 0.0025 for the models of employment, sales volume and productivity variables, respectively. This shows that there is no specific spatial pattern in the errors conforming to the geography of the urban environment under study. There is no evidence of a spatial lag in the specification that would be omitted, nor is this study omitting any predictor that would show a strong spatial tendency.

5. Discussion

Forecasting economic development is critical as it informs both policy makers and business owners about the future prospects of different areas. By knowing with reasonable confidence about the future in advance, policy makers will be able to prepare if they need to investigate influential factors to set suitable policies for managing declining trends or encouraging growing trends in specific neighborhoods. Effective local forecasts may also be leveraged by public agencies to anticipate the needs for physical infrastructure (such as transportation) and human infrastructure (such as schools, day care centers or health facilities) in or near neighborhoods with fast evolving trends. Business owners and real estate developers can use forecasting to learn about locations with higher potential economic returns. Therefore, adopting more accurate forecasting models will result in higher efficiency in the urban economy. Performing the forecasts at fine geographic resolutions, such as neighborhoods, assists policy makers, community development managers, business owners, or real estate

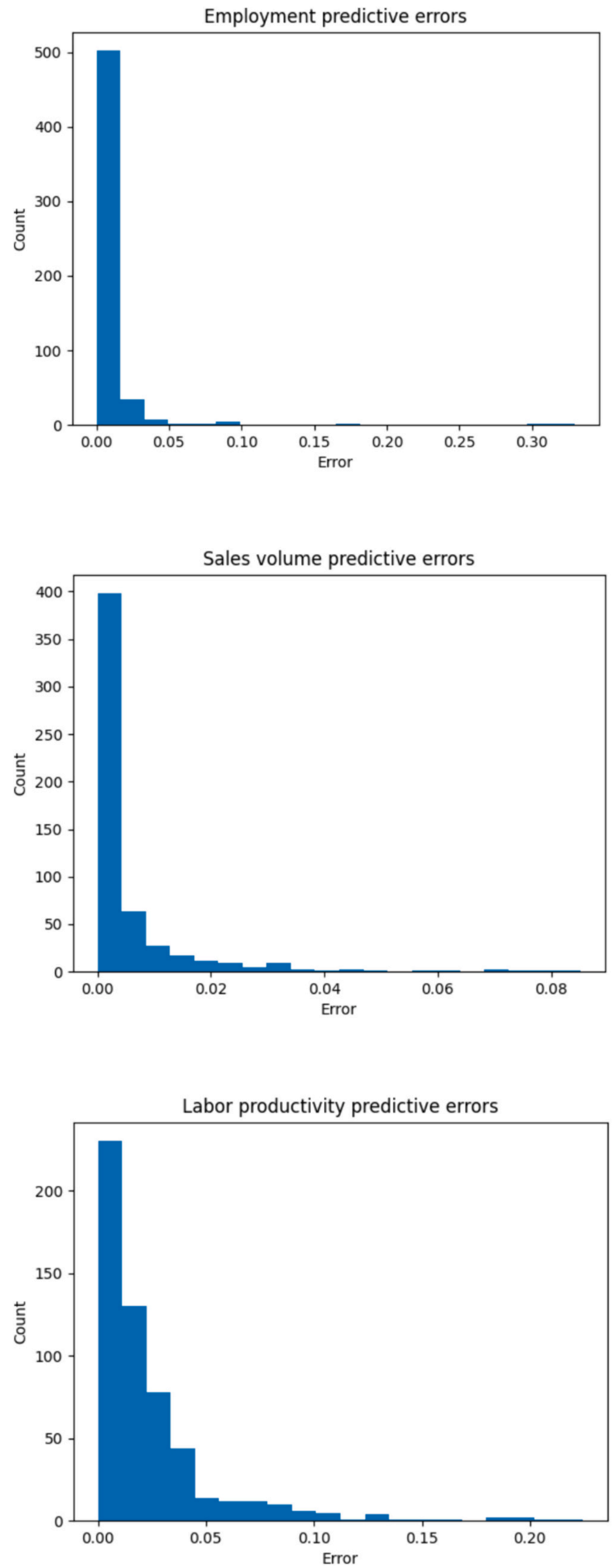

Fig. 3. Statistical distribution of normalized predictive errors (normalized by the range of observed values) for the year 2016.

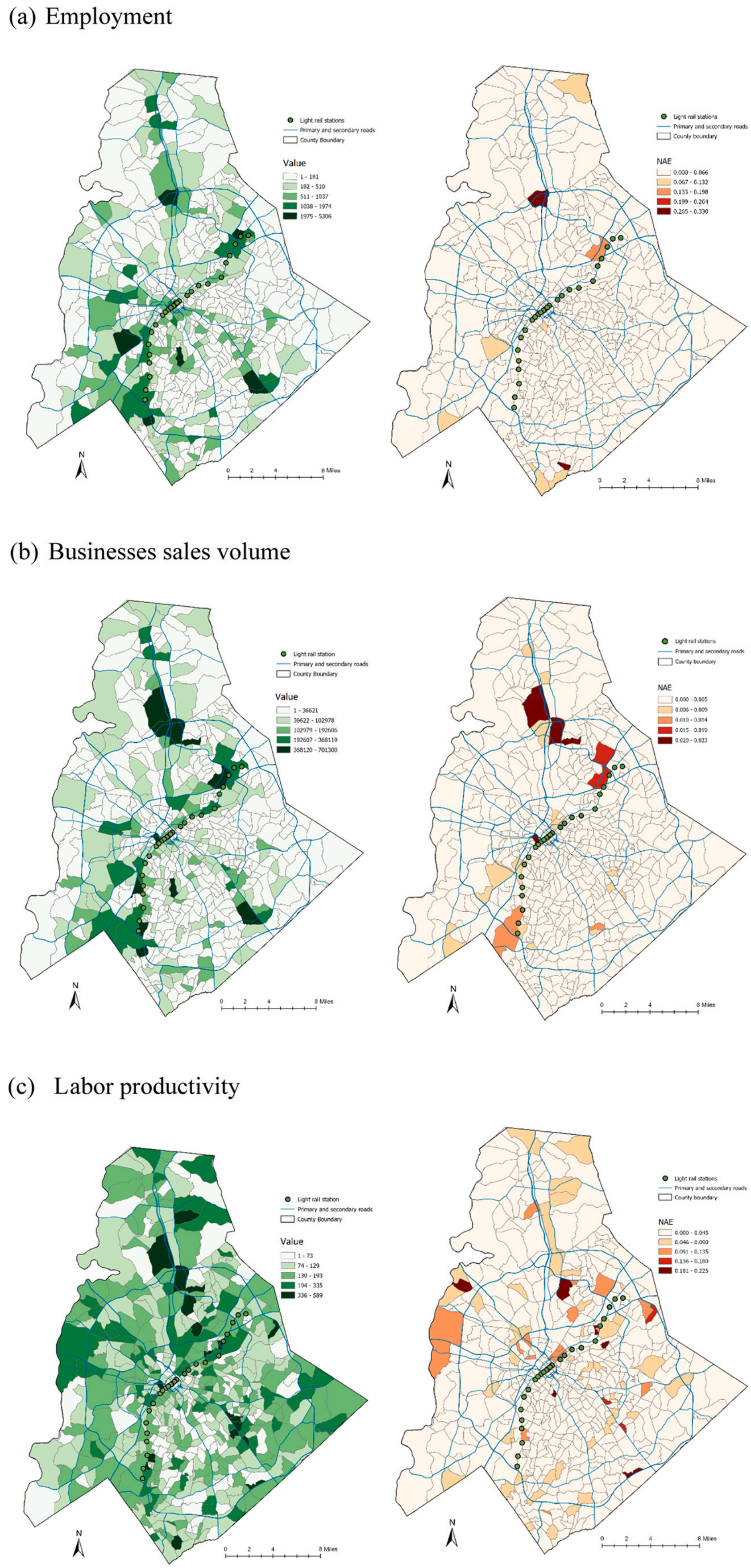

Fig. 4. Observed values and model predictive errors (absolute error normalized by range) in 2016.

developers in making more accurately targeted decisions.

High geographic granularity is essential for a number of reasons. First, as explained through one of the well-known problems in geographic analysis, namely the modifiable areal unit problem (Openshaw, 1979), scale of study is a prominent determinant in analysis outcomes. Second, and on a related note, as building blocks of the regional economy, there is considerable heterogeneity in neighborhood economic and demographic characteristics that need to be studied. Third, even for regional economic strategies, interconnections between neighborhoods and regions, in addition to the heterogeneity in neighborhoods, need to be considered (Weissbourd, 2006). This point becomes more significant in policy making as different interventions and policies are required for different geographic granularities. Fourth, recent advancements in data availability at the fine spatiotemporal resolutions of neighborhoods annually make the analysis of disaggregated economic dynamics now feasible.

This study makes several significant contributions to literature. First, it covers the research gap of neglecting the essentiality of forecasting in urban economics where causal inference studies have been predominant (Credit, 2018; John, 1996; Lv et al., 2021; Martínez & Viegas, 2009; Mohammad et al., 2013; Schmidt et al., 2022). Second, it takes a state-of-the-art data-driven approach and uses advanced techniques including deep learning, feature extraction, and dimensionality reduction using a large number of variables from a variety of data sources. Regarding the theoretical implications, the proposed method alleviates the issues in the conventional methods used in economic research, such as statistics and econometrics. Utilizing deep learning and increasingly available high-resolution data, future researchers will have fewer limitations that come from traditional statistical models, including multicollinearity, non-linearity, and normality. Third, it successfully generates small-area predictions, which have been particularly wicked problems to resolve. Hence this opens the door to a range of new opportunities for place-based policy formulation and forecast that had become dormant. Service industries consist of a considerable share of the economy, and their dynamics vary across different types, mainly whether they are in the group of business services or non-business services. As a result, in this study, three industries were selected and studied as non-business services together. These industries include retail, accommodation and food services, and other services with 2-digit NAICS codes of 44–45, 72, and 81, respectively.

In this study, three economic indicators of employment, business sales volume, and labor productivity were selected as outcome variables. To forecast these economic outcomes, a series of covariates interrelated with economic dynamics were used in a multivariate model, including the sociodemographic, economic, built environment, and real estate investments. Long Short-Term Memory (LSTM) Recurrent Neural Networks (RNN) was employed to handle the nonlinearities in the relationships. A spatiotemporal LSTM-RNN model was developed to forecast the outcomes for future years. In order to evaluate the model through out-of-sample testing, the dataset covering all block groups in Mecklenburg County, NC, for the years 2002–2018 was split into training and testing sets over temporal dimensions with proportions of 80 % and 20 %. The last three years are used for testing the model performance. Our model learns the relationships based on all census block groups together, and forecasts are done for each block group individually. Spatial dependence was not found to bias the prediction results. Our results indicate strong performance of the model for all three outcome variables of employment, business sales volume, and labor productivity, with MAE values of 0.235, 0.206, and 0.164 as normalized by the mean of observed values; and MAE values of 0.011, 0.012, and 0.034 as normalized by the range of observed values.

For future work considerations, having longer historical data is expected to enhance the model's predictive performance. Depending on data availability, we will consider using longer historical data of economic outcomes for future work as well as a broader study area that naturally fits the extent of the metropolitan region. Furthermore, using information on the businesses' costs as an additional input would help us develop an economic indicator well suited to the profitability of businesses. As the most useful information from the business perspective, future profits can be studied in future work.

The ability to predict the future of economic outcomes in the county, particularly in competitively growing areas such as the mentioned transit corridors, would be tremendously informative to investors and business owners to pinpoint areas with higher returns on investments. In addition, the fine geographic resolution of census block groups helps them to find their desired location in competitive areas with higher precision. The proposed spatiotemporal modeling framework has the capability of predicting outcomes in other industries as well.

6. Conclusions

In conclusion, although the causal relationships between different social, environmental, and economic factors are tremendously important, the literature must pay more attention to the future forecasts of economic outcomes in addition to causal inference. The ability to predict the prospective outcomes benefits a variety of stakeholders, such as the public sector, urban planners and designers, business owners, and real estate investors, as well as the local residents.

In this study, a spatiotemporal deep learning model was developed that predicts the economic outcomes, including employment, business sales volume, and labor productivity for three years at the fine geographic resolution of block groups.

Our results indicate strong performance of the model for all three outcome variables of employment, business sales volume, and labor productivity, with MAE values of 0.235, 0.206, and 0.164 as normalized by the mean of observed values; and MAE values of 0.011, 0.012, and 0.034 as normalized by the range of observed values. Normalization is done to have a better understanding of the error values given the wide range of observed values across all block groups. Looking into the results in Figs. 3 and 4, only a very small number of block groups exhibit large errors. The majority of those cases are block groups with large observed values and it is natural to have larger predictive errors for larger observed values. One potential explanation for the high predictive error of employment and sales in the block groups along a corridor to the north of the city center is the opening of the last section of a major peripheral highway (I-485 intersecting with the I-77 highway) in 2013. Such significant change in the spatial structure of the city may trigger sudden jumps from historical trends. Additionally, as shown in the map and histogram of predictive errors for productivity (Figs. 3 and 4), there are relatively more block groups with large errors than with the other two outcome measures. This is, to a large extent, due to higher uncertainty in this metric. The productivity metric is calculated based on two inputs of employment and sales volume, each of which brings some level of uncertainty and noise. As a result, the productivity metric becomes noisier and more challenging to predict.

As shown by maps of all three indicators, high economic outcomes are located along the current light rail corridors as well as proposed additional mobility corridors. These transit corridors were planned to integrate transportation and land use by enhancing accessibility to a mixture of land uses, including single-family and multifamily residential, retail and office (Charlotte-Area-Transit-System, 2019; City of Charlotte, 1998). One of the main objectives of transit-oriented developments along these corridors has been to promote economic development by bringing the origin-destination of home, work, and retail trips closer together. As shown in maps of observed values (Fig. 4), Mecklenburg County has large values of employment and sales volume along the transit corridors. In predictive modeling, it is normally expected to have larger errors for larger values. The maps of errors (Fig. 4) indicate that the model developed in this article performs very well in this respect as there are not many block groups with large errors along the transit corridors.

This study has a few limitations. First, the time series is constrained by data that do not extend beyond 2002. From the economic perspective, 20 years of annual data is a considerable amount; however, from the data analytical perspective, having a longer history makes forecasting more accurate. In addition to the temporal constraints discussed, this study is further confined by its geographical limitations, primarily due to data availability being restricted to one county rather than encompassing surrounding areas as well. One of our considerations for future work is to expand the study area to the broader metropolitan area to capture more interactions and dynamics within a functional urban boundary rather than within the bounds of the jurisdiction of Mecklenburg County. Considering the majority of interactions and dynamics happening within Mecklenburg County, this study is not problematic; however, incorporating interactions with neighboring counties may give a better and more comprehensive representation of the interdependent workings of the urban region. Unfortunately, surrounding counties do not make some spatial data such as built environment shapefiles or building permit data available as Mecklenburg County does. Therefore, this study calls for attention to the importance of data integration and to building appropriate infrastructures to supply data for a functional region such as metropolitan areas, considering the matter of consistency. As a second future work consideration, learning about the relative impact of covariates on economic outcomes will be tremendously insightful for policymakers. In practice, knowing what variables are more influential in predicting economic growth in small areal units, along with the direction of impacts, will be very useful. Machine learning techniques such the SHAP values method can be effective at studying the ranking of influential variables along with the magnitude and direction of the impacts (Hatami et al., 2023).

To sum up, this research provides value to both the research community and practitioners. It adds value to the research community by recommending the usage of cutting-edge AI techniques to resolve the shortcomings of traditional econometric models, including limitations in multicollinearity, high-dimensionality, and complex relationships. It provides tremendous benefits to practitioners by giving insights into the future of economic indicators at the fine geographic resolution of neighborhoods, and by industry type. By knowing the future dynamics in advance, practitioners including policy makers and investors are able to identify areas that are poised to grow or at the brink of decline and adjust their actions accordingly.

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Faizeh Hatami: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ahad Pezeshk Poor:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Jean-Claude Thill:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

Authors have no conflict of interest to declare.

Data availability

Data will be made available on request.

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